

Practice Exercises

07:00

Exercise 2 - SQL Fundamentals

08:00

(Aggregate Functions & Grouping)

09:00

1) Write a SQL query to find the total number of employees in the company.

10:00

SELECT COUNT(*) AS total-employees

11:00

FROM employees;

12:00

Total-employees

10

13:00

2) Write a SQL query to find the total salary paid to all employees in the IT Department

14:00

15:00

SELECT SUM(salary) AS total-it-salary

15:30

FROM employees

16:00

WHERE department = 'IT'

16:30

17:00

Total-it-salary

220 000

18:00

NOVEMBER 2021

Su	Mo	Tu	We	Th	Fr	Sa
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				



Start KM

End KM

Total KM



07:00 *Write a SQL query to calculate the average salary of employees in the HR department*

08:00

08:30

09:00

09:30

*SELECT AVG(salary) AS average-hr-salary
FROM employees
WHERE department = HR;*

10:00

10:30

11:00

11:30

average-hr-salary
49500

$(48000 + 51000) \div 2$
49500

12:00

12:30

13:00

13:30

Write a SQL query to find the highest and lowest salary in the company

SELECT

14:00

14:30

15:00

15:30

*MAX(salary) AS highest-salary
MIN(salary) AS lowest-salary
FROM employees;*

16:00

16:30

17:00

17:30

Highest - Salary	Lowest Salary
62000	48000

18:00

"Belief in yourself is more important than endless worries of what others think of you. Value yourself and others will value you. Validation is best that comes from within." - Ngugi wa Thiong'o, Dreams in a Time of War

OCTOBER 2021						
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3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

Start KM	End KM
Total KM	

"Claiming our right to have our human dignity and safety protected and respected in pursuit of a decolonised, quality, public education"

08-00

100

11:00

1998

14:00

6:00

7:00

8:00

20

2021

OCTOBER WEDNESDAY

- World Statistics Day

Time	City	Number of emp
07:00	Chicago	3
08:00	Houston	1
09:00	Los Angeles	2
09:30	New York	2
10:00	San Francisco	2

① Write a SQL query to group employees by department, calculate the average salary in each department, and order the results in descending order of average salary.

SELECT department,
AVG (Salary) AS average_salary
FROM employees
GROUP BY department
ORDER BY average_salary DESC;

Department	Average salary
Finance	\$9 500
IT	\$8 000
Marketing	\$2 500
HR	\$9 500

"Your own actions are a better mirror of your life than the actions of all your enemies put together."
- Ngugi wa Thiong'o, Wizards of the Crow

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31						

Start KM _____ End KM _____ Total KM _____

"Claiming our right to have our human dignity and safety protected and respected in pursuit of a decolonised, quality, public education"

08:00 (b) write a SQL query to find department where the total salary paid exceeds 100 000

08:00

09:00
SELECT department,
SUM(Salary) AS total-Salary
FROM employees

10:00

GROUP BY department
Having SUM(Salary) > 100 000

11:00

11:30

department	total-Salary
IT	220 000
Finance	119 000
Marketing	105 000

12:00

12:30

13:00

13:30

14:00

14:30

15:00

15:30

16:00

16:30

17:00

17:30

18:00

(a) write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order

SELECT City
Count(*) AS number-of-employees

FROM employees

GROUP BY City

HAVING Count(*) > 1

ORDER BY number-of-employees DESC;

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Start KM End KM Total KM






